

JESSE WHEELER

Assistant Professor
Department of Mathematics and Statistics
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Academic Positions

Assistant Professor of Statistics *2026–Present*
Department of Mathematics and Statistics
Utah State University, Logan, UT.

Assistant Professor of Statistics *2025–2026*
Department of Mathematics and Statistics
Idaho State University, Pocatello, ID.

Education

PhD in Statistics *2020–2025*
University of Michigan, Ann Arbor, MI.
Thesis Title: “Innovations in Likelihood-Based Inference for State Space Models”
Thesis Advisor: Edward Ionides

B.S. in Mathematics, Statistics, Minor in Computer Science *2016–2020*
Utah State University (USU), Logan, UT.
Graduated as valedictorian of the College of Science, class of 2020

Research

Interests

Time Series Analysis
Mechanistic Models
Statistical Practice and Data Science
Modeling Dynamic Ecological Systems
Fisheries sciences and management

Peer Reviewed Publications

Wheeler, J., Ionides, E. L. (2025). Revisiting Inference for ARMA Models: Improved Fits and Superior Confidence Intervals. *PLOS ONE*, 20(10): e0333993. doi:10.1371/journal.pone.0333993
Bretó, C., **Wheeler**¹, J., King, A., Ionides, E. L. (2025). panelPomp: Analysis of Panel Data via Partially Observed Markov Processes in R. *The R Journal*, 17, 180-199. doi:10.32614/RJ-2025-009.
Ionides, E. L., Ning, N. and **Wheeler, J.** (2024). An iterated block particle filter for inference on coupled dynamic systems with shared and unit-specific parameters. *Statistica Sinica*, 34, 1241-1262. doi:10.5705/ss.202022.0188.

¹Corresponding Author

- Wheeler, J.**, Rosengart, A., Jiang, Z., Tan, K., Truetle, N., Ionides, E. (2024). Informing policy via dynamic models: Cholera in Haiti. *PLOS Computational Biology*, 20(4), e1012032. doi:10.1371/journal.pcbi.1012032
- Wagstaff, J., Bean, B., **Wheeler, J.**, Maguire, M., Sun, Y. (2024). Adaptive Mapping of Design Ground Snow Loads in the Conterminous United States. *Journal of Structural Engineering*, 150(1), 04023193. doi:10.1061/JSENDH.STENG-12396
- Wheeler, J.**, Bean, B., Maguire, M. (2022). Creating a Universal Depth-to-Load Conversion Technique for the Conterminous United States Using Random Forests. *Journal of Cold Regions Engineering*, 36(1), 04021019. doi:10.1061/(ASCE)CR.1943-5495.0000270
- White, T., **Wheeler, J.**, Lindstrom, C., Christensen, R., Moon, K. (2021). GPS-Denied Navigation Using SAR Images and Neural Networks. *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Toronto, ON, Canada, 2021, pp. 2395-2399, doi:10.1109/ICASSP39728.2021.9414421.

Non-Refereed Publications

- Bean, B., Maguire, M., Sun, Y., Wagstaff, J., Al-Rubaye, S., **Wheeler, J.**, Jarman, S., Rogers, M. (2021). The 2020 National Snow Load Study. *Utah State University*. doi:10.26077/200k-pr86.
- Ionides, E. L., **Wheeler, J.** (2024). Review 2: “Efficacy, Public Health Impact and Optimal Use of the Takeda Dengue Vaccine.” *Rapid Reviews Infectious Diseases*. doi:10.1162/2e3983f5.1f0cb1f4.

Working Papers

- Wheeler, J.**, Abkemeier, A. A., Ionides, E. L. (2026+). Iterating marginalized Bayes maps for likelihood maximization with application to nonlinear panel models. *Submitted to the Journal of the American Statistical Association, Theory and Methods* arXiv:2511.17438.
- Yang, B., **Wheeler, J.**, King, A., Duffy, M., Ionides, E. L. (2026+). Mechanistic Models for Panel Data: Analysis of Ecological Experiments with Four Interacting Species. *Submitted to the Annals of Applied Statistics*. arXiv:2506.04508.
- Omotowa, O., Ankarberg, L., Gist, K., **Wheeler, J.**, Nies, M. (2026+). Caring for Older Adults in Nursing Homes: Attitudes and Willingness of Racial Minority Undergraduate Nursing Students. *In preperation for The Educational Gerontology journal*.

Seminars, Presentations, and Workshops

- 2026** Idaho State University, Math / Stat department research colloquium.
- 2026** American Fisheries Society, Idaho Chapter Annual Meeting (poster). Idaho Falls, ID.
- 2025** Idaho State University, invited colloquium speaker in Biology Department.
- 2025** Epidemics 10: 10th International Conference on Infectious Disease Dynamics (poster). San Diego, CA.
- 2025** The Geological Society of America, Connects 2025 (Poster). San Antonio, TX.
- 2024** Modelling and inference for pandemic preparedness. Isaac Newton Institute, Cambridge, England.

- 2024** Michigan Student Symposium for Interdisciplinary Statistical Sciences. Ann Arbor, MI.
- 2023** Bayes Comp 2023, Satellite Event: Bayesian Inference of Epidemics. Levi, Finland.
- 2023** Models of Infectious Disease Agent Study (MIDAS) Network Annual Meeting (Poster). Atlanta, GA.
- 2022** JSM 2022. Washington D.C.
- 2020** National Conference on Undergraduate Research. Online (COVID-19).
- 2020** USU Student Research Symposium (Poster). Logan, UT.
- 2020** Utah Conference on Undergraduate Research (Poster). Logan, UT.

Research Consulting

Statistical Consultant, NIH R16 (SuRE-First) grant application with Dr. Solomon Zeleke.

Healthy Idaho CTR-D Pilot Project Awards. *NIH funded project.* Part of NIH-grant supported team of biostatisticians to review and consult on NIH developemental and pilot grants. Listed as a key collaborator for the CTR-D submission by Dr. Solomon Zeleke.

Grants and Funding

Development of a statistical framework for the evaluation of management strategies for freshwater fisheries. *Oak Ridge Associated Universities: Ralph E. Powe Junior Faculty Enhancement Award.* Submitted January, 2026.

Awards

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| StatsForward Fellowship , The American Statistical Association. | <i>September 2025</i> |
| Best Oral Presentation , MSSIS 2024, University of Michigan.
<i>Award amount: \$200</i> | <i>2024</i> |
| Rackham Graduate Student Research Grant , University of Michigan
<i>Award amount: \$2500</i> | <i>2024</i> |
| Honorable Mention NSF GRFP | <i>2022</i> |
| Rackham Merit Fellowship , University of Michigan
<i>Award amount: Three years of PhD student funding.</i> | <i>2020–2025</i> |
| Valedictorian , USU College of Science | <i>2020</i> |
| URCO Grant , USU
<i>Award amount: \$1000</i> | <i>2020</i> |

Teaching

Curriculum Development

Revision of **Math 4450**, *Idaho State University*. Redesign course that had not been taught by any current faculty. Course material is available on accompanying course website: https://jeswheel.github.io/4450_f25/.

Revision of **Math 4451**, *Idaho State University*. Redesign course that had not been taught by any current faculty. Course material is available on accompanying course website: https://jeswheel.github.io/4451_s26/.

Idaho State University

2025–present

Math 4451/5451, Spring 2026 (Mathematical Statistics II)
Math 4450/5450, Fall 2025 (Mathematical Statistics I)
Math 3350, Fall 2025, Spring 2026 (Calculus-based introduction to statistics)

University of Michigan

2020–2025

Stats 604, *Lab Instructor* (Statistical Practice)
Stats 531, *Office hours / Grading* (Modeling and Analysis of Time Series Data)
Datasci 415, *Lab Instructor* (Introduction to Statistical Learning)
Stats 306, *Lab Instructor* (Introduction to Statistical Computing)
Summer Math Boot Camp (Calculus, Linear Algebra, Probability, Computing)
Workshop *Developing a personal academic website* (Student Seminar Series)
Tutor, *Master’s level Rackham Merit Fellows* (Probability and Regression)

SISMID (2025). Instructor for a short course on *Simulation based inference for Epidemiological Dynamics* at the Summer Institute in Statistics Modeling in Infectious Diseases (SISMID). Emory University, Atlanta, Georgia.

SISMID (2022). Instructor for a short course on *Simulation based inference for Epidemiological Dynamics* at the Summer Institute in Statistics Modeling in Infectious Diseases (SISMID). University of Washington, Seattle.

Utah State University

2017–2020

Math 0995, *Recitation Leader* (Remedial Algebra)
Math 1210, *Recitation Leader* (Introductory Calculus)
Math 1220 *Recitation Leader* (Calculus II, sequences and series)
Stats 1040, *Recitation Leader*, (Introductory Statistics, non-Calculus based)

Undergraduate Research Projects Mentored

Hayley Torrez. TBD. 2026.
Spencer Paternoster. Restricted Maximum Likelihood estimators for ARMA models. 2025.
Peter Yang. Applications of Partially Observed Markov Processes with phylogenetic data. Honors thesis, 2025.
Weizhe Sun. Model Based Inference of Stochastic Volatility via Iterated Filtering. Honors thesis, 2024.
Zuyuan Han. Signature Methods in Variance Swap Pricing. Honors thesis, 2023.

Bo Yang. Analysis of Panel Data via Mechanistic Models in a PanelPOMP Framework. Honors thesis, 2023.

Kevin Tan and Noah Treutle. On the Transmissibility of Cholera During the 2010–2019 Haiti Cholera Epidemic. Research project, 2022.

Awards

Outstanding Graduate Student Instructor Team Award. *2025*

Outstanding Undergraduate Recitation Leader *2019*

USU Mathematics and Statistics Department

Service

Computing Club Committee Member, UM Statistics Department *2022–Present*

Committee Chair *2023–Present*

Invited Judge, The American Statistical Association, Statistics in Epidemiology Section: 2026 SIE Early Career Paper Competition.

Peer Reviews: PLOS Computation Biology, Nature Communications, Rapid Reviews Infectious Diseases, Bulletin of Mathematical Biology, Scientific Reports, Communications Health

President, USU Data Science Club *2019–2020*

Software

Author and maintainer for the R package **arima2**, available on CRAN. As of October 1, 2024, this package has been downloaded more times than 79.7% of all packages on CRAN since it was first published (October 5, 2023).

Primary contributor and current maintainer of the R package **panelPomp**, available on CRAN.

Core developer of the Python package **pypomp**, available on PyPI.

Contributor to open source R packages **pomp** and **spatPomp**, both available on CRAN.

Paid Consulting

Kasiska Division of Health Sciences, Idaho State University. *2025–2026*

Campfire Interactive, time series modeling. *Nov 2024–Jan 2025*